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ITAI 1371
Final Project Outline

Machine Learning Final Project Outline

Interest Area

Music / Media

Step 1 — Start With You

I am interested in music and how machine learning can be used to keep a listener in the right mood or "vibe" by learning from the songs they liked in the past.

Step 2 — Turn Your Interest Into ML Project Ideas

1. Can ML predict whether a user will like a new song using their previous likes and skips?

2. Can ML classify songs into different vibe categories such as chill, energetic, sad, or hype?

3. Can ML detect a user's current music vibe from the songs they recently listened to?

4. Can ML recommend the next song that best matches a user's vibe?

Chosen Idea:
Can ML recommend the next song a user will like based on the songs they previously liked or interacted with positively?

Step 3 — Learn What the Best Work is Doing (State of the Art)

Research Paper

Paper: "A Survey of Music Recommendation Systems and Future Perspectives"

1. What are they trying to solve?
   They are trying to solve the problem of recommending songs that match a user's taste.

2. What data did they use?
   They used music listening history, ratings, song metadata, and user behavior data.

3. What ML method did they use?
   Collaborative filtering, content-based filtering, and hybrid recommendation systems.

4. What is one important result?
   Hybrid recommendation systems usually perform better because they use both song features and user behavior.

Trusted Article / Blog

Source: Spotify Engineering Blog

1. What are they trying to solve?
   Spotify is trying to keep users listening longer by recommending songs that match their mood and preferences.

2. What data did they use?
   Spotify uses listening history, likes, skips, playlists, and song features such as tempo and energy.

3. What ML method did they use?
   Recommendation systems, collaborative filtering, and deep learning.

4. What is one important result?
   Spotify found that personalized recommendations increase user engagement and listening time.

Step 4 — Look at 3 Big Companies

Company	What are they building?	How does ML help?	Interesting Detail
Spotify	Personalized playlists like Discover Weekly and DJ	ML recommends songs based on listening history and similar users	Spotify's AI DJ talks to the user and explains song choices
Apple Music	Song and playlist recommendations	ML studies what users like, skip, and save	Apple Music creates custom stations based on one song or artist
YouTube Music	Personalized mixes and recommendations	ML learns from watch history and liked songs	YouTube Music combines both video and music behavior

Step 5 — Shrink Your Topic to Match This Course

1. What dataset will you use?
   I will use a public music recommendation dataset from Kaggle, such as the Spotify

Tracks Dataset or Million Song Dataset.

2. What is your input (features)?
   The input features could include:

- Song tempo
- Energy
- Danceability
- Genre
- Artist
- Previous user likes or skips
- Number of times a song was played

3. What is your output (target)?
   The target will be whether the user is likely to like the next song.

- 1 = User likes the song
- 0 = User does not like the song

4. What is your algorithm type?
   This is a recommendation and classification problem.

5. Which models will you try?

- Logistic Regression
- K-Nearest Neighbors
- Decision Tree
- Random Forest

I expect Random Forest or KNN to work best because they can find patterns between similar songs and similar users.

Step 6 — Abstract

Music recommendation systems are important because users want songs that match their mood and personal taste. Many music apps use machine learning to keep users listening by recommending songs they are likely to enjoy. Researchers and companies such as Spotify, Apple Music, and YouTube Music use listening history, song features, and user behavior to build recommendation systems. This project will use a public music dataset to build a machine learning model that predicts whether a user will like a song based on their previous listening behavior and the characteristics of the song. Several machine learning models will be compared, including Logistic Regression, Decision Tree, Random Forest, and K-Nearest Neighbors, to see which model gives the best recommendations.